

KnowServe Corp. — Engineering Department

System Architecture Overview

The architecture of KnowServe Corp.'s intelligent document management platform is designed to ensure efficiency, scalability, and maintainability across multiple integrated services. The system is based on a modular microservice approach, allowing each component to operate independently while communicating through secure REST APIs and asynchronous event queues. At the core lies a robust FastAPI backend that orchestrates all data flow between user-facing components and internal services. PostgreSQL serves as the relational database backbone, maintaining transactional consistency for user, document, and departmental data. For vector-based search and AI retrieval, a Chroma vector database integrates seamlessly with the backend, enabling contextual document embeddings and intelligent semantic lookup capabilities. Caching and message brokering are handled via Redis, providing rapid response times and resilience against latency spikes during peak loads. The system's file management integrates directly with Google Drive through a secure OAuth-based service, ensuring both cloud storage redundancy and controlled access permissions at the department level. Each deployment is automated using Docker and CI/CD pipelines configured in GitHub Actions. The pipeline automatically builds, tests, and deploys new service versions to the Kubernetes cluster, minimizing manual intervention and reducing downtime. Logging and monitoring are centralized using Prometheus and Grafana dashboards, allowing real-time visibility into system performance and error patterns. The architecture prioritizes security from the ground up: JWT-based authentication ensures secure API access, role-based permissions control departmental boundaries, and encrypted data storage safeguards sensitive documents. Future enhancements will include horizontal scaling strategies and AI-driven load balancing, ensuring KnowServe's infrastructure continues to evolve alongside the growing needs of its users. This document serves as a high-level technical overview for new engineering team members and stakeholders who seek to understand how KnowServe Corp. achieves reliability, performance, and modular expansion across its ecosystem.